Dynamic Asset Allocation in a Regime Switching Model of the Economy

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Abstract

Identifying economic regimes is useful in a world of time-varying risk premia. Regime shifts present a big challenge to traditional strategic asset allocation, demanding a more adaptive approach. In the presence of time varying investment opportunities, portfolio weights should be adjusted as new information arrives. Regime-switching models can match the tendency of financial markets to change their behavior abruptly and the phenomenon that the new behavior often persists for several periods after a change. They are well suited to capture the stylized behavior of many financial series including skewness, leptokurtosis, volatility persistence, and time-varying correlations. The following research builds on this empirical evidence to develop a quantitative framework for regime-based asset allocation. It investigates whether regime-based investing can effectively respond to changes in financial markets at the portfolio level in an effort to provide better long-term results when compared to more static approaches. We have employed a two-state Gaussian Hidden Markov Model and the Expectation-Maximization algorithm to identify the market regimes based on several macro-economic indicators and show that the ensuing regime factor is relevant in building a multi asset portfolio that outperforms static asset allocation strategies. Finally, these forecasts exhibit significant out of sample predictability that ultimately translates into considerable utility gains in a Mean-Variance and Mean-CVaR portfolio strategy.

Keywords: Macroeconomic indicators, Regime Switching, Hidden Markov Model, Expectation-Maximization, Forward-Backward Algorithm, Dynamic Programming, Mean-Variance, Mean-CVaR

This is a very interesting project, and is extremely well conceived and extremely well written. The project has a feeling of completeness that is rare for projects in the course.

A+

I really have no suggestions for making the project better. The project is excellent.

Introduction

The behavior of financial markets changes abruptly. While some changes may be transitory, the changed behavior often persists for many periods. Considering the ongoing turmoil spurred by the Covid-19 Financial Crisis (CFC), one may have the impression of a paradigm shift in investors' assessment of risky assets. The mean, volatility, and correlation patterns in stock returns, for example, changed dramatically at the start of, and continued through the global financial crisis (GFC) OF 2007-2008. Similar regime changes, some of which can be recurring (recessions vs expansions) and some of which can be permanent (structural breaks), are prevalent across a wide range of financial markets and in the behavior of many macro variables. The observed regimes in financial markets are closely related to the phases of business cycle. Regime changes present a big challenge to traditional strategic asset allocation, demanding a more adaptive approach. Moreover, the time varying behavior of risk premiums, volatilities, and correlations have important implications for risk-management.

In a risk-on/risk-off world, regime switching methods appeal as a natural remedy for identifying the current state of the market and adjusting one's portfolio strategy accordingly. The following research examines whether the regime-based asset allocation can effectively respond to financial regimes in an effort to provide better long-term results when compared to static approaches. It is detected that a bull market is characterized by low asset correlations and a bear market is characterized by high market correlations. We have employed univariate regime switching models based on measures of market performance, market turbulence, inflation, economic growth, and unemployment to identify economic regimes. We explore whether a strategy that dynamically scales its exposure to specific risk premia in accordance with these regimes could perform superior to a strategy with constant risk premia exposure. It turns out that using information from the above-mentioned indicators, we are able to predict NBER recessions and that can be used to successfully navigate the prevalent economic regimes by means of a dynamic asset allocation strategy.

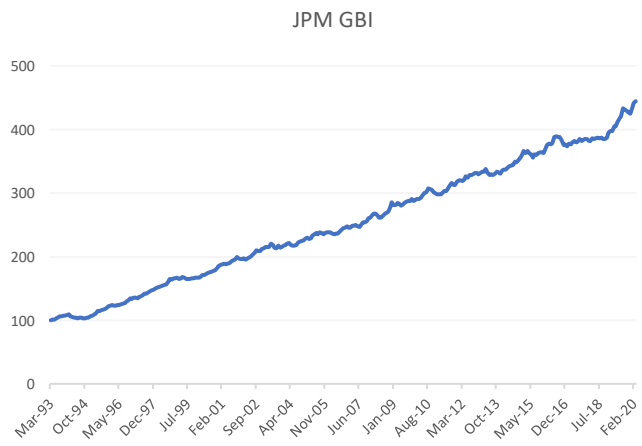
nicely written

The following report has been divided into several sections. In section 1, we give a brief overview about the dataset and the various asset classes that has been used for the analysis of this project and their sources. In section 2, we briefly explain the theory behind Markov chains and the Hidden Markov Model and how Expectation-Maximization algorithm, and its variation can be effectively used to identify different states of the economy (regimes) at every point in time. In section 3, we run through the various variables and macro-economic indicators that have been used to identify the market regime at every point in time, both in-sample and out of sample. We then compute the average probabilities of being in each of the two regimes along with the probabilities of transition from one state to the other at every point in time. In section 4, we present the idea of dynamic asset allocation and use the results obtained thus far to construct a Mean-Variance optimized portfolio. We also backtest our regime based dynamic asset allocation strategy out of sample and compare the strategy's performance with a static asset allocation and a Mean-Variance optimized portfolio that disregards the regime-based information. Based on the statistics, we can confidently say that the regime-based asset allocation significantly outperforms a comparable benchmark and the equity market index and promises utility gains

in the long term. In section 5, we perform a similar analysis by constructing a regime-based Mean-CVaR asset allocation strategy and again compare its performance with static asset allocation strategies. In section 6, we conclude the analysis by summarizing the steps taken and present our recommendations. Lastly, the Appendix section goes through some deeper and more technical explanations of the algorithms used in this project along with some minor examples to show their efficacy.

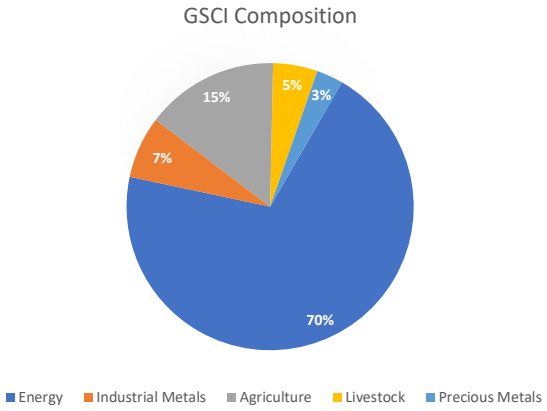
I. Data

The purpose of this project is to compare the performance of a regime-based asset allocation strategy under realistic assumptions to a strategy in a similar environment but excluding the regime information. Thus, we will show the efficacy of the regime specific information in portfolio construction. It will be examined whether the outperformance can be achieved when there is no risk-free asset, but rather the possibility for diversification by holding a portfolio of assets that represent broad asset classes. The asset classes considered are limited to stocks, bonds, and commodities to keep it simple yet complex enough for diversification possibilities to arise. The data includes a U.S. equity index (S&P 500), a U.S. Government Bond Index (JP Morgan Government Bond Index), and a Commodity Index (Goldman Sachs Commodity Index). Monthly closing prices covering the 27-year period from 1993 to 2020 are considered. The data prior to 2000 will be used for in-sample analysis and estimation, while the 20 years from 2000 to 2020 will be used for out of sample analysis and development of backtest statistics. The in-sample data is special in that it includes both a market rally period and the 2000 dot com bubble, thus evidently representing two regimes to learn from. The out of sample data can be used to see if the model accurately predicts the 2008 Global Financial Crisis (GFC) and 2020 Covid-19 Financial Crisis (CFC). Everything is measure in USD as seen from a US investor's point of view. In this way, there will be no need to consider currency risk. We would like to mention that the JPM GBI (bond index) and, GSCI (Commodity Index) historical data have been downloaded from Bloomberg via the Bloomberg Terminal. The S&P500 (equity index) historical data has been downloaded from Yahoo Finance. The statistical software R is used for all data analysis, modelling and portfolio optimization.









	Equity	Commodity	Bond
Equity	1	0.3	-0.15
Commodity	0.3	1	-0.19
Bond	-0.15	-0.19	1

Correlation matrix based on monthly returns from 1994 to 2020

	Equity	Commodity	Bond
Mean	0.006	0.00058	0.00466
Std dev	0.042	0.065	0.009
Skewness	-0.75	-0.57	0.016
Excess Kurtosis	1.35	2.25	0.099
Ann Sharpe Ratio	0.514	0.030	1.745

Summary statistics based on monthly returns of the 3 assets

From the correlation matrix, we can see that bonds are negatively correlated with both equity and commodities. Hence, one can expect to increase their allocation in bonds when the equity market crashes or during recessionary regimes. That is precisely our motivation to build a regime switching model, so that investors can advantage from large gains by allocating in equities during market rally or expansionary regimes and protect themselves from large drawdown by investing in bonds during recessionary regimes.

From the summary statistics based on monthly returns of the 3 assets, we can see that the annualized Sharpe Ratio of the JPM GBI is higher as compared to SP500 and GSCI. This is because of the extremely low volatility of bond returns. The negative skewness of equities and commodities suggest that the left tail of the distribution of returns of the two assets are fatter than the right tails, i.e. investors suffer from excess losses during market crash than they gain during market rally. We believe that the regime-switching model could help investors to protect themselves from these heavy left tails.

Along with the asset returns data, we have downloaded several macro-economic indicators data as mentioned earlier from the FRED website (https://fred.stlouisfed.org/). They have been explained in more detail in the forthcoming sections. We shall be using these indicators to identify regimes at every point in time by employing a two-state gaussian Markov-switching model, which will then be used to construct efficient portfolios to identify asset allocations. The rebalancing period that we assume for the analysis is one month.

II. Markov based Regime Switching Model

We will begin by describing a basic Markov-switching model and why such a model is often superior to a naïve approach to identifying regimes. We will start with an example from outside of finance. Imagine that someone wears a heart rate monitor and we receive this person's heart rate data at one-minute intervals. While this person is sleeping, we observe a low average heart rate with low volatility. And, when the person is awake, we observe a sudden rise in the heart rate of this person with high volatility. The objective of this imagination is to conclude whether the person is awake or asleep, without seeing the person. The heart rate data follows a Markov process; at any point in time, a "state" or "regime" (asleep or awake) generates observations (heart rates) from a specific distribution. The regime changes over time – hence the term Markov switching process.

Let the initial probability ($t = 0$) of being in regime i is given by,

$$Pr(X_0 = i) = p_i$$

Where X_1 is the first regime in the Markov chain. The transition probability matrix is denoted by $\mathbf{\Gamma}$. Within the transition probability matrix, the elements γ_{ij} denote the probability of a transition into regime j from regime i as follows:

$$\mathbf{\Gamma} = \begin{pmatrix} \boldsymbol{\gamma_{11}} & \boldsymbol{\gamma_{12}} \\ \boldsymbol{\gamma_{21}} & \boldsymbol{\gamma_{22}} \end{pmatrix}$$

$$\gamma_{ij} = P(X_t = j | X_{t-1} = i)$$

Where t denotes time. Therefore, over time the Markov chain is in either regime $X_t = 1$ or $X_t = 2$. Each regime generates observations Y_t that are consistent with a given distribution π_i. For a discrete distribution (with a finite set of possible values for Y_t), the equation

$$\pi_i = P(Y_t = s | X_t = i)$$

Indicates the current regime X_t dictates the probability that Y_t, will take specific value s. The distribution π_i may take many forms. For example, we could assume that Y_t is normally distributed with regime-specific mean μ_i and standard deviation σ_i and recast the preceding equation as

$$Y_t \sim N(\mu_i, \sigma_i)$$

Where *N(.)* represents the normal distribution probability density function.

Similar to the heart rate example above, we assume that financial markets too follow a Markov switching model. There are two states of the market – Expansion and Contraction – and these states are unobserved directly. However, we can estimate these states based on several observed data that follow a two-state regime based gaussian distribution.

$$Y_t = \mu_{S(t)} + \sigma_{S(t)} W(t)$$

Where Y_t are the observed values, *S(t)* represents the regime estimated at time *t*, $\mu_{S(t)}$ is the mean of the series *Y* during the regime *S(t)*, $\sigma_{S(t)}$ represents the standard deviation of the series *Y* during regime *S(t)*, and *W(t)* is a standard normal random variable. We can also simplify the transition probability matrix as:

$$\mathbf{\Gamma} = \begin{pmatrix} \mathbf{1-p} & \mathbf{p} \\ \mathbf{q} & \mathbf{1-q} \end{pmatrix}$$

Here, *p* represents the probability of transitioning from regime 1 to regime 2 or $\boldsymbol{\gamma_{12}}$, *1-p* represents the probability of persistence in regime 1 or $\boldsymbol{\gamma_{11}}$, q represents the probability of transitioning from regime 2 to regime 1 or $\boldsymbol{\gamma_{21}}$, and *1-q* represents the probability of persistence in regime 2 or $\boldsymbol{\gamma_{22}}$. Besides the transition probabilities, we also need to estimate the probability of being in regime 1 and regime 2 given the observed series Y_t.

$$\eta = P(S(0) = 1)$$

$$1 - \eta = P(S(0) = 2)$$

Finally, we bundle all the parameters that we need to estimate based on the observed series Y_t.

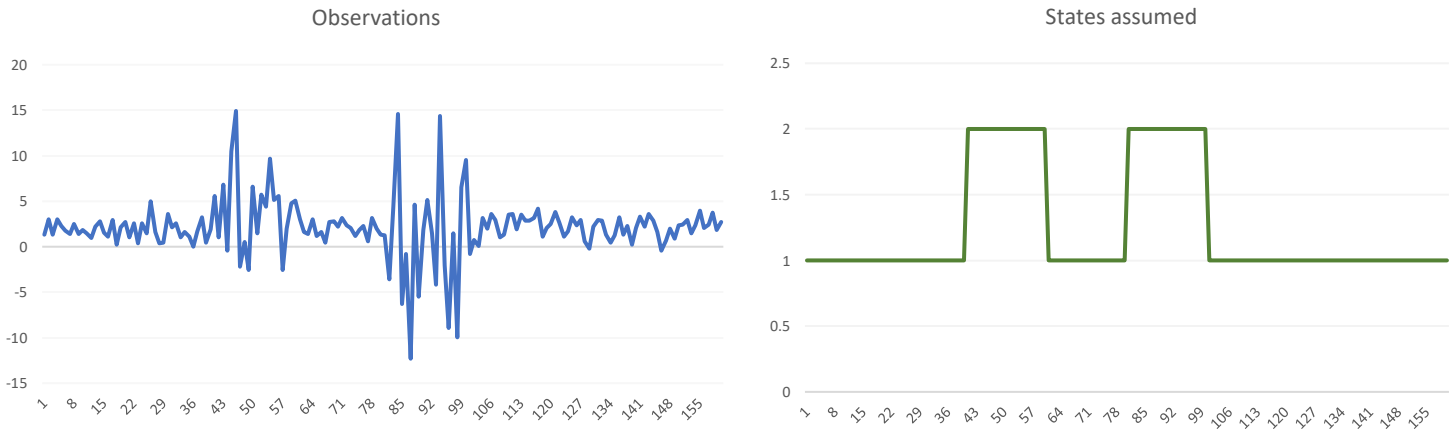
$$\theta^{MSM} = (\mu_1, \mu_2, \sigma_1, \sigma_2, \eta, p, q)$$

Where *MSM* is short-hand for Markov Switching Model, 1 and 2 in the subscript represent parameters for regime 1 and regime 2 respectively.

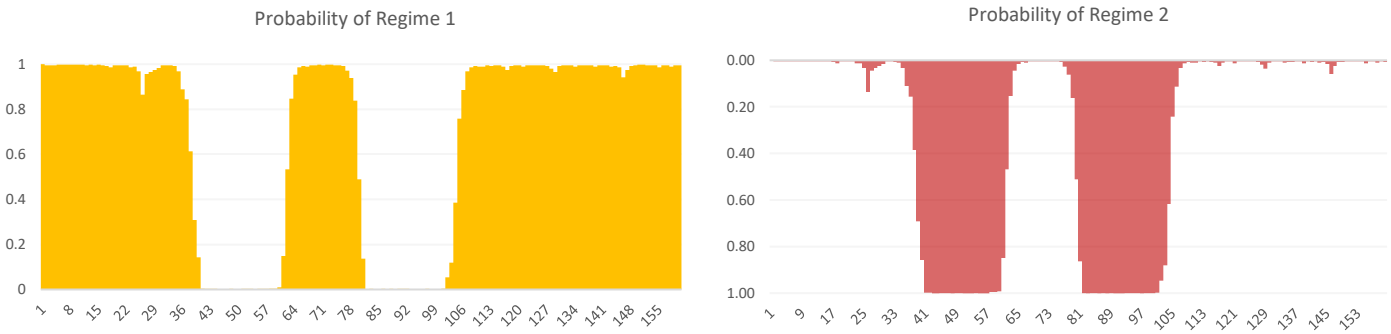
To show some of the advantages of Markov Switching models, we built a simple 'toy' model with known parameters and simulated sample observations for Y_t. We made the following assumptions for this experiment.

1. In Regime 1, Y_t is normally distributed with a mean of 2 and sigma of 1.
2. In Regime 2, Y_t is normally distributed with a mean of 4 and sigma of 6.

We generate the following sequence Y_t and model the series by employing a two state Gaussian Markov Switching Model as explained above and estimate the model's parameters θ via the Expectation-Maximization algorithm which is based on the Maximum Likelihood Estimation.



Simulated Markov Switching Process



Output from the MSM model

The smoothed probabilities from the MSM model is shown above. With a threshold of 0.9 for Regime 2, the accuracy of the predicted regimes is 98.125% and we get only 3 misclassifications. The transition probability matrix that we get is

$$\mathbf{\Gamma} = \begin{pmatrix} 0.906 & 0.093 \\ 0.005 & 0.995 \end{pmatrix}$$

This suggests that the persistence in Regime 1 is over 90% while the persistence in Regime 2 is over 99%. This example just shows us the power of using Regime switching models and is a huge motivation to explore how Markov Switching Models perform on financial datasets.

III. Parameter Estimation in a Regime Switching Model

The parameters of the Hidden Markov Model described above are usually estimated using the Maximum Likelihood Estimation (MLE) method. This process consists of 3 steps:

1. Evaluating the likelihood of the observations Y_t given the parameters.
2. Estimating the model parameters θ that maximize the likelihood of the observed data
3. Inferring the most likely sequence of states $S(t)$

The first step is addressed by a dynamic programming technique, the Forward-Backward Algorithm, the second task is carried out by a variation of the Expectation-Maximization algorithm known as the Baum-Welch algorithm and the third task can be solved either locally or globally. Local decoding means to determine the most likely state at each time *t,* and global decoding will help us determine the most likely sequence of states. Since our aim is to develop a trading strategy which investors can use to allocate weights to assets in real time, we will use the method of local decoding to estimate state *S(t)*. Viterbi algorithm is generally used for Global decoding when one wishes to determine the most likely sequence of states given all the data Y_T up to time T.

There are several research papers describing the technique in detail used to identify the states in an HMM. So, in this report, instead of diving into the technical details, we offer a simple description of the technique for a two-state model governed by simple normal distributions. We have employed a similar model and structure throughout the remaining research as well. Moreover, the same principles can be extended to estimate more complex regime models. Unfortunately, there is no closed form solution to this problem because the states underlying the model are

unobservable. We must resort to using an algorithm to effectively 'learn' the best set of parameters by taking an initial estimate of the transition matrix and improving the estimate in each successive iteration. The Baum-Welch algorithm provides an efficient way to solve this problem. We want to solve the set of parameters that maximizes the probability of observing the data given the parameters.

$$\theta^* = \max_{\Theta} P[y_1 y_2 \dots y_T|\theta]$$

Where Θ is the set of all choices of parameters θ and $P[y_1 y_2 \dots y_T|\theta]$ represents the probability of observing the Y_t series that we are provided, conditional on a given set of parameters.

Following the Baum-Welch algorithm, we begin by picking arbitrary values for the transition matrix **Γ**. Assuming for a moment that these are indeed the true transition probabilities, we can compute the precise probability of each sequence of unobservable states $S(t_1), S(t_2), \dots . S(t_n)$. The overall likelihood of observing the data is, therefore, the probability weighted sum – or Expectation – of the data likelihood of each possible regime sequence. We can then solve for the two distributions and a new set of transition probabilities that jointly maximizes the expected likelihood (hence the term "Expectation Maximization" algorithm). We feed these new transition probabilities back into the algorithm and repeat the process, refining our estimates of the transition matrix until the algorithm converges and the likelihood has been maximized. Each iteration is guaranteed to produce a new transition matrix that contributes at least as much likelihood as the previous estimate.

We shall be using the term smoothed probabilities when we refer to the probability of any observed data point Y_t being in any regime *i.* The smoothed probability is the product of forward probability, $\alpha_i(t)$ and backward probability $\beta_i(t)$.

$$\alpha_t(i) = P(Y_1 = y_1 \dots ., Y_t = y_t, S_t = i|\theta)$$

$$\beta_t(i) = P(Y_{t+1} = y_{t+1} \dots, Y_T = y_T|S_t = i, \theta)$$

$$\alpha_t(i)\beta_t(i) = P(Y_T = y_T, S_t = i|\theta)$$

Thus, the likelihood of the observed data y_T, can be evaluated as the sum over all *m* states:

$$P(Y_T = y_T|\theta) = \sum_{i=1}^{m} \alpha_t(i)\beta_t(i)$$

Finally, the smoothed probability of Y_t being in Regime *i*, is calculated as:

$$p_t(i) = \frac{\alpha_t(i)\beta_t(i)}{\sum_{i=1}^{m} \alpha_t(i)\beta_t(i)}$$

We suggest the reader to go through some of the past research on Regime switching models mentioned under the References section for a more thorough and technical explanation of the above algorithms.

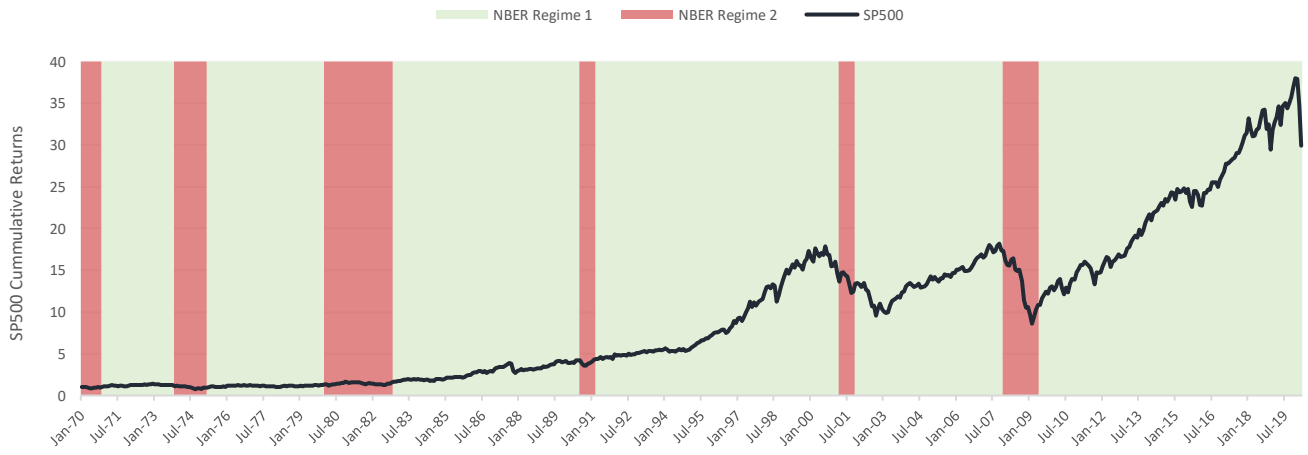
IV. Interpreting Economic Regime variables

The regimes we identified in our empirical study are based on the time series for equity market returns, financial market turbulence, inflation, economic growth, and unemployment. This list is not meant to be exhaustive, however, each of these variables captures a different aspect of the economic activity and provides a useful and concise dataset for our analysis.

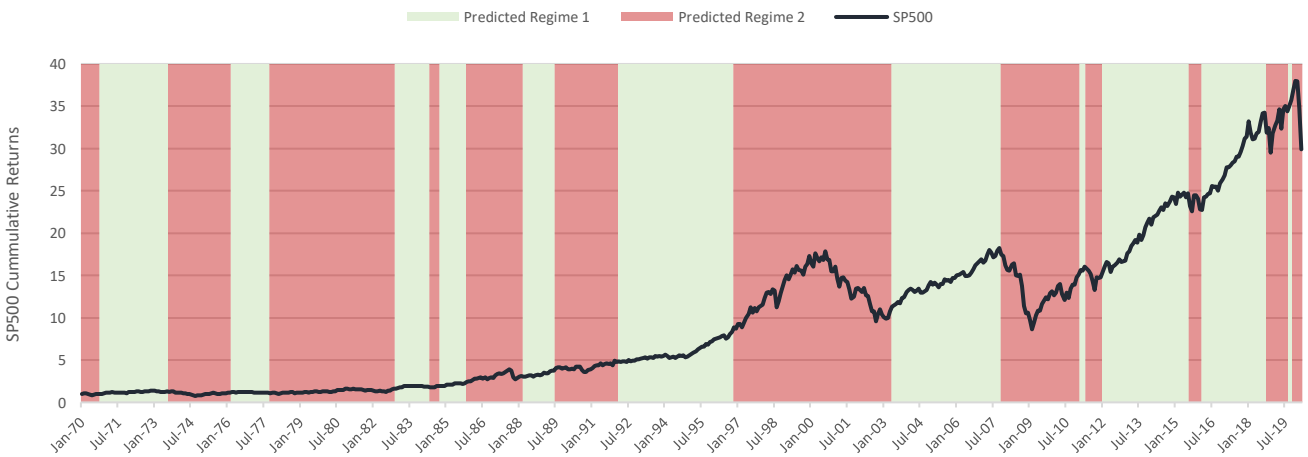
We will present both in-sample and out of sample estimates of regime probabilities and transition probabilities. The in-sample period is 1970-2000 for most of the observed series Y_t and out of sample period is 2001-2020. We will also compare the regime predictions from the MSM model with the recession dates as decided by NBER. This will help us to keep track of the performance of our model at each step and utility of each economic variable. These dates, and data for macroeconomic indicators have been downloaded from the FRED website as mentioned earlier. Finally, we will be combining the smoothed probabilities from each univariate model and compute an equally weighted average at every point in time as the final probabilities $p_i(t)$ of both regimes, and similarly for transition probabilities.

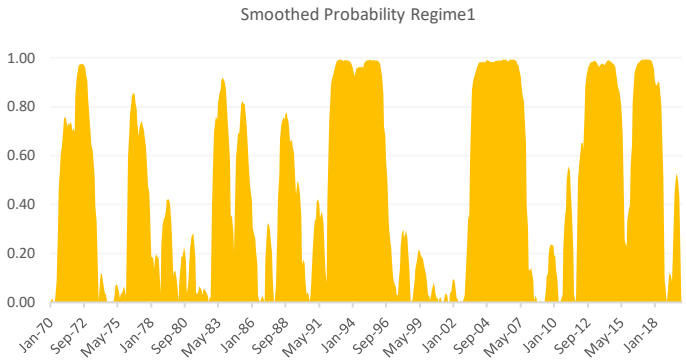
1. Equity Returns or SP500 Returns

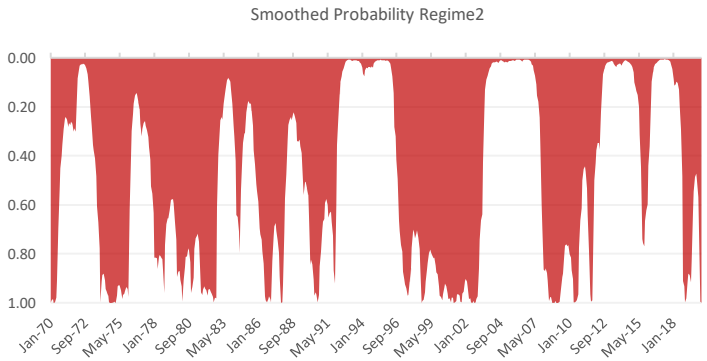
We have used monthly returns of S&P500 index as a proxy for U.S. equity returns.



Performance of SP500 and Regimes based on NBER recession dates







To calculate the threshold probability for any data y_t to be classified as the event regime (Regime 2), we looked at the average smoothed probability of Regime 2 during the dates mentioned by NBER as the recession regime and found that to be ~ 0.7. So, we fixed a threshold of 0.7 for smoothed probability of Regime 2, for observed data y_t to be classified as Regime 2. The other parameter estimates are:

$$\mu = (0.01, -0.003)$$

$$\sigma = (0.027, 0.055)$$

$$\Gamma = \begin{bmatrix} 0.96 & 0.04 \\ 0.05 & 0.95 \end{bmatrix}$$

The parameters suggest that average monthly return during Regime 1 is 1% while the average monthly return during regime 2 is -0.3%, and monthly volatility in returns during Regime 1 is 2.7% while monthly volatility in Regime 2 is much higher to 5.5%. These estimates seem reasonable and align with our assumption of Regime 2 being more volatile with more challenging investment conditions than Regime 1 (lower μ). It is also interesting to note that both the persistence numbers are larger than 50% and that persistence of event regime is lower than that of the non-event regime. This can be interpreted intuitively as the natural reaction to turbulence is to reduce portfolio risk and the natural reaction to recession is to stimulate the economy, both of which should eventually make conditions more stable.

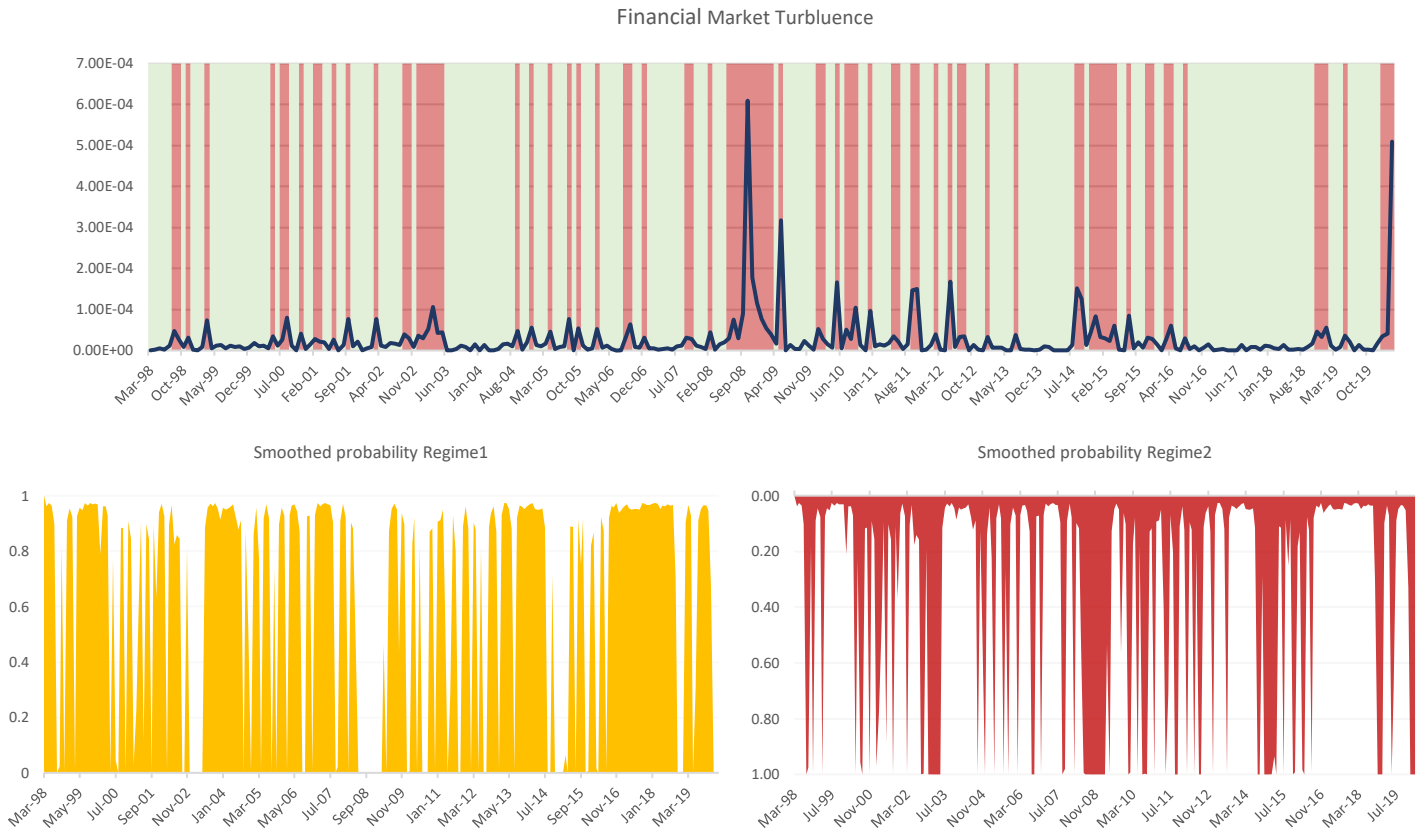
2. Financial Market Turbulence

Financial market turbulence can be defined as a condition in which asset prices behave in an uncharacteristic fashion given their historical pattern of behavior, including extreme price moves, decoupling of correlated assets, and convergence of uncorrelated assets. Specifically, we measured financial turbulence using the following multivariate distance measure (sometimes referred as the squared Mahalanobis distance).

$$d_t = (y_t - \mu)\Sigma^{-1}(y_t - \mu)^T$$

where y_t is the vector of asset returns for period *t*, μ is the sample average vector of historical returns, and Σ is the sample covariance matrix of historical returns.

We compute the monthly financial market turbulence at every time *t*, using past 5 years of monthly returns of the three asset classes under consideration – equities, commodities, and bonds.

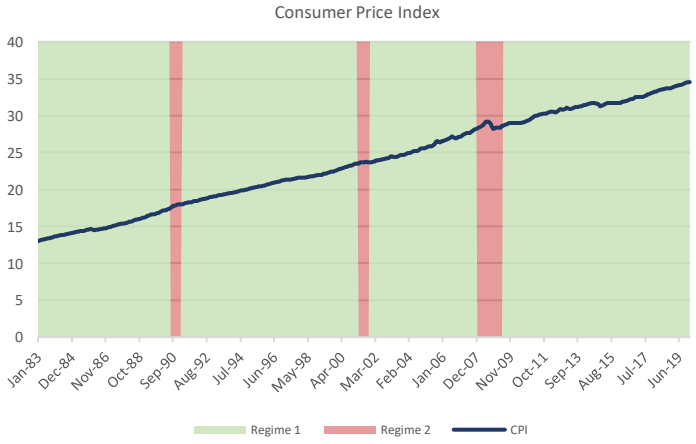


Smoothed Probabilities and predicted regimes based on Market Turbulence

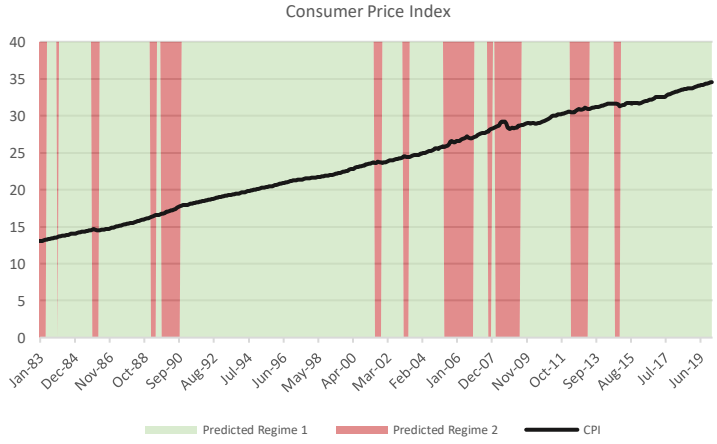
3. Inflation

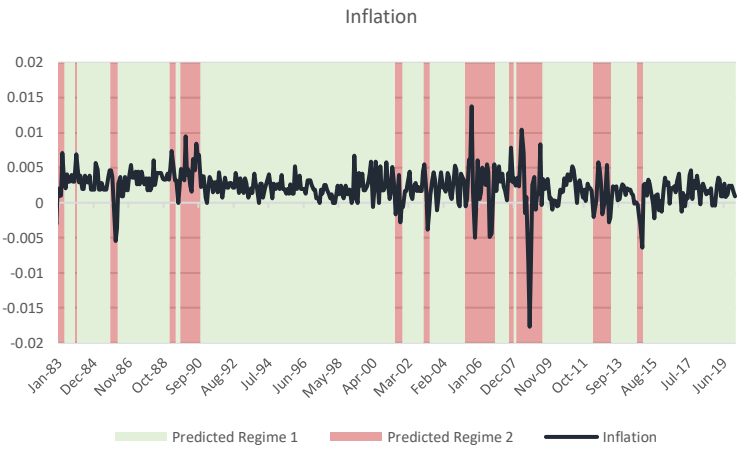
We measure inflation using monthly percentage changes in the seasonally adjusted U.S. Consumer Price Index for All Urban Consumers from the Federal Reserve Economic Data website.

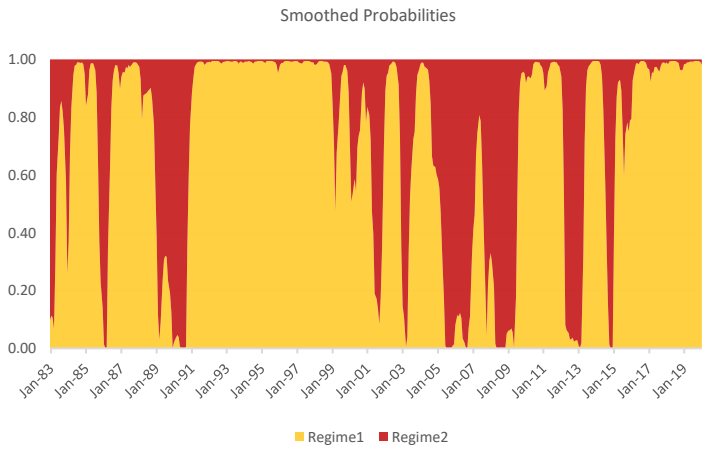
Regimes based on NBER dates



Regimes predicted by MSM with a threshold of 0.7





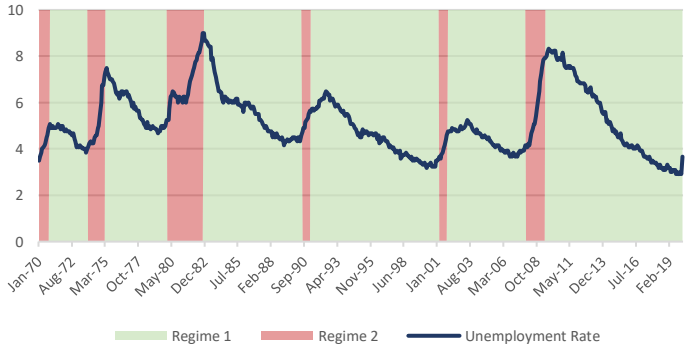


An interesting result to note from the table below is that high inflation regimes are highly persistent. This could likely be because of the self-reinforcing cycle of higher wages and higher prices.

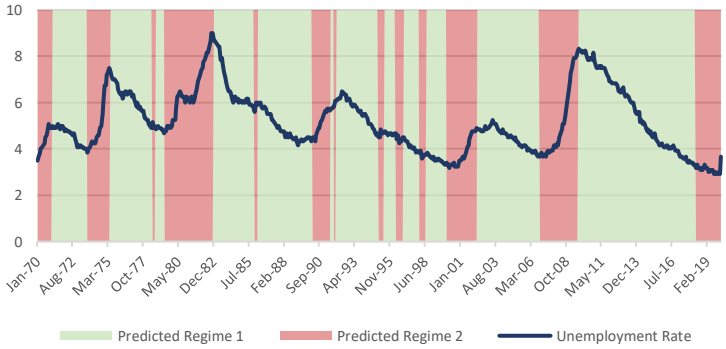
4. Unemployment

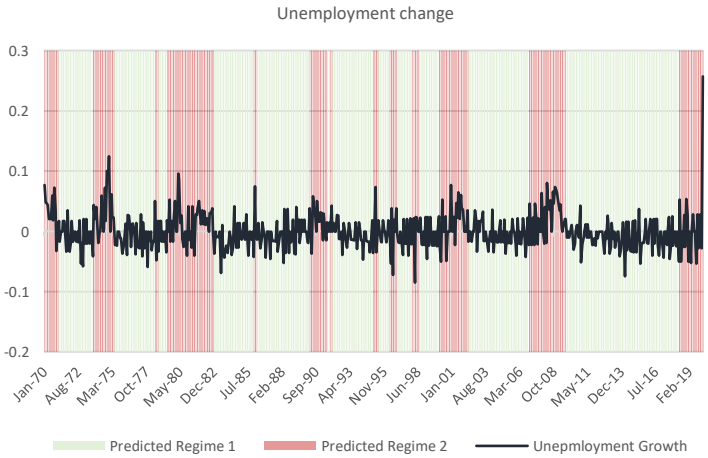
The change in the unemployment rate is measured as the monthly percentage change in the seasonally adjusted US unemployment rate. Again, the data is downloaded from Federal Reserve Economic Data website.

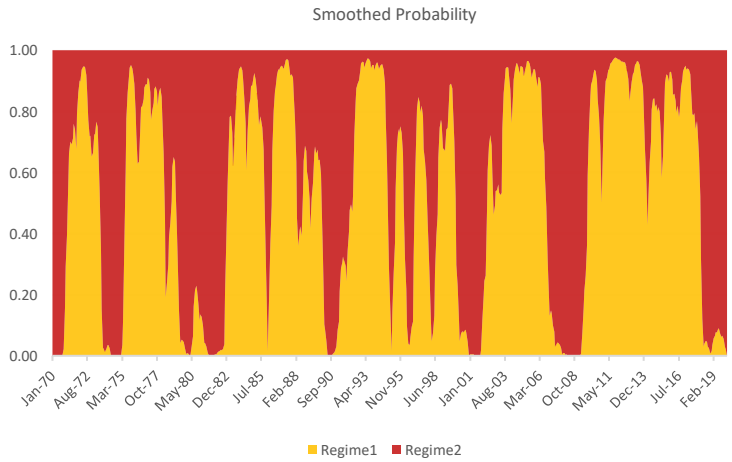
Regimes based on NBER dates



Regimes predicted by MSM with a threshold of 0.7







Markov Switching Model: In-Sample Estimation Results

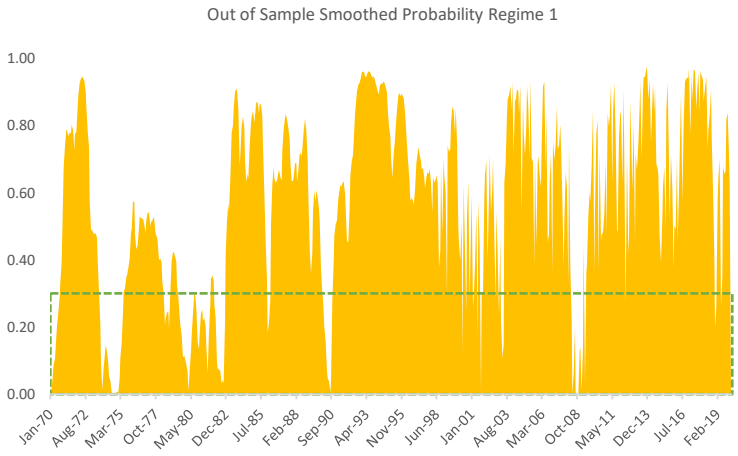
	Regime 1 (normal)			Regime 2 (event)		
	Persistence	Mu (annualized)	Sigma (annualized)	Persistence	Mu (annualized)	Sigma (annualized)
Equity returns	96%	12%	9.35%	95%	-3.60%	19.05%
Turbulence	60%	0.07	0.05	67%	0.61	0.85
Inflation	97%	2.40%	0.49%	91%	5.28%	1.50%
GNP growth	91%	1.09%	0.84%	68%	-0.14%	0.96%
Unemployment change	89%	-9.80%	6.80%	97%	12.90%	12.50%

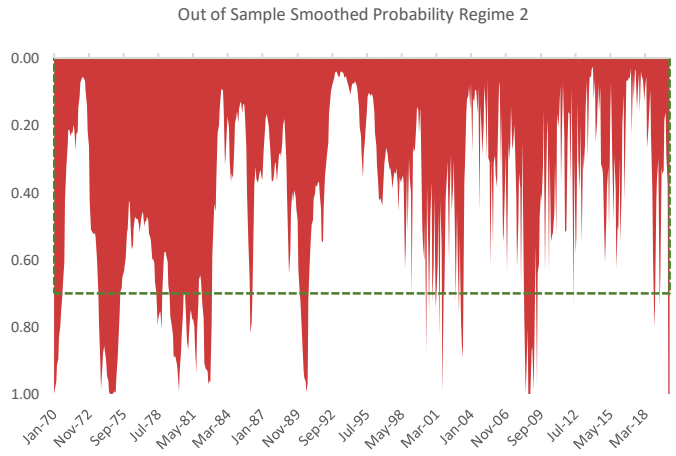
Note: Persistence is defined as the estimated transition probability of staying in the current regime

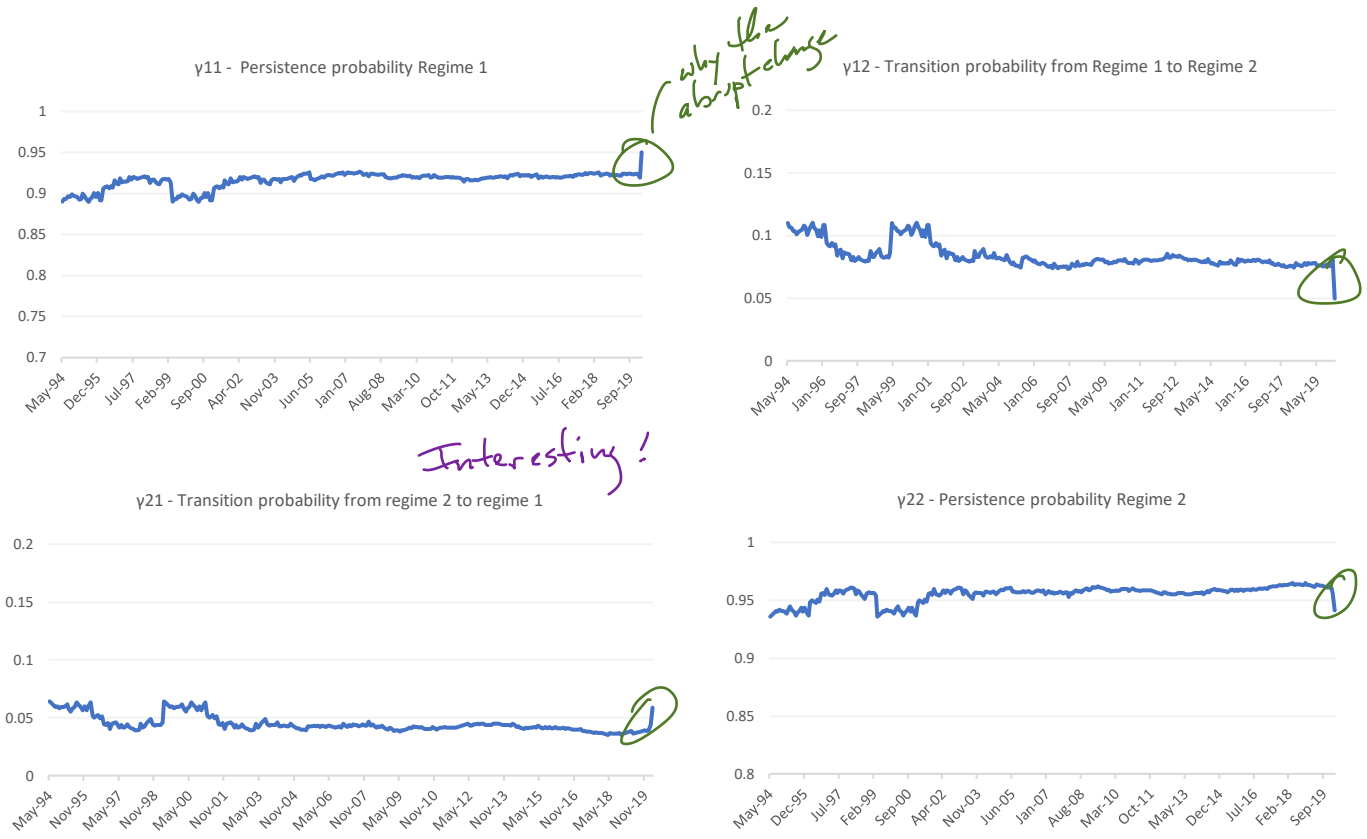
V. Out of Sample Analysis

Our in-sample results showed that Markov-switching models succeeded in partitioning historical data in a meaningful way, but they did not reveal whether a dynamic process would outperform a static process out of sample. To test out of sample performance, we designed the following experiment. Each month, we implemented these steps:

1. We calibrated our Markov switching model using a growing window of data from inception up to that point in time.
2. We compute the final smoothed probability of each regime as the average of the 5 macroeconomic indicators' out of sample smoothed probability. Similarly, we calculate the average transition probabilities γ_{ij}.
3. Consistent with our in-sample analysis, we assign a threshold of 0.7 for the smoothed probability of Regime 2 to classify that point in time *t,* as Regime 2.
4. Depending on which regime the economy is at current time *t,* we compute the expected return and covariance matrix of returns between the three asset classes to construct a long-only portfolio with mean-variance optimization.







Out of Sample average transition probabilities from the MSM model

VI. Dynamic Asset Allocation – Mean Variance Portfolio Optimization

We aim to build a Mean-Variance dynamic portfolio by anticipating crises and market rally. We calculate the expected return and covariance matrix of the three assets – Equities, Bonds, and Commodities by utilizing the out of sample transition probabilities and the past 24 months sample returns. We compare the performance of this dynamic strategy with a static mean-variance portfolio, a strategy which simply allocates weights within the portfolio every month, with similar assumptions, i.e. past 24 months sample and the three assets.

According to standard Markowitz theory, the expected return and risk of a portfolio is represented as

$$E[R_p] = \mu^T x$$

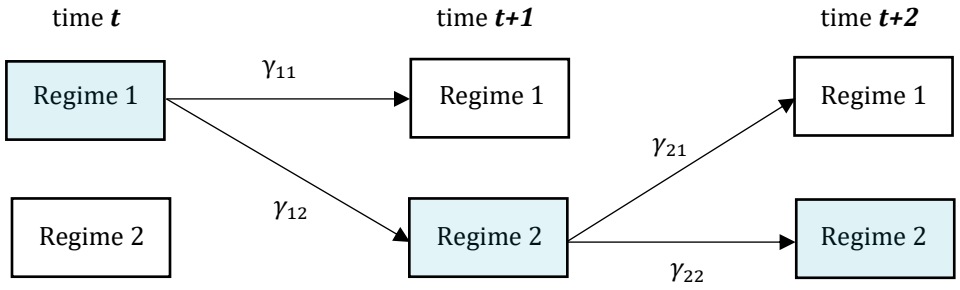
where, $\mu = [\mu_1 \mu_2 \ldots . \mu_N]^T$ and $x = [x_1 x_2 \ldots x_N]$

$$Var(R_p) = x^T \Sigma x$$

where Σ is the nXn variance-covariance matrix of the n assets.

For our research, to make use of the regime information and anticipating the possibility of transitions between regimes at every point in time, we slightly modify the calculation of expected return and variance of the portfolio. We calculate conditional expected return at time *t+1* given an estimated Regime *i* at time *t* as a probability weighted average return

within the two Regimes with samples from the past 24 months for each regime. Similarly, we calculate the conditional variance of returns of the portfolio as probability weighted average variance of past sample returns of the three assets. The following illustration will help to better explain the idea of conditional expectation and conditional variance of portfolio returns given estimated regime at time *t*.



Regime Realizations (shaded)

	time *t*	time *t+1*	time *t+2*
Conditional Expected Return	$\gamma_{11}\mu^w(1) + \gamma_{12}\mu^w(2)$ $= E[R_p^{t+1}](1)$	$\gamma_{21}\mu^w(1) + \gamma_{22}\mu^w(2)$ $= E[R_p^{t+2}](2)$	$\gamma_{21}\mu^w(1) + \gamma_{22}\mu^w(2)$ $= E[R_p^{t+3}](2)$
Conditional Variance	$\gamma_{11}[\sigma^w(1)]^2 + \gamma_{12}[\sigma^w(2)]^2 + \gamma_{11}\gamma_{12}[\mu^w(1) - \mu^w(2)]^2$ $= \Sigma^w(1)$	$\gamma_{21}[\sigma^w(1)]^2 + \gamma_{22}[\sigma^w(2)]^2 + \gamma_{21}\gamma_{22}[\mu^w(1) - \mu^w(2)]^2$ $= \Sigma^w(2)$	$\gamma_{21}[\sigma^w(1)]^2 + \gamma_{22}[\sigma^w(2)]^2 + \gamma_{21}\gamma_{22}[\mu^w(1) - \mu^w(2)]^2$ $= \Sigma^w(2)$

Here, $\mu^w(1)$ is the sample mean of past returns (24 months in our implementation) of the three assets during Regime 1. So, we sub-select the most recent 24 months when the market has occurred in Regime 1 based on the out of sample estimate and calculate the sample mean of monthly returns in those 24 months. Similarly, $\mu^w(2)$ is the sample mean of past returns of the three assets during Regime 2. And, $\sigma^w(1)$ is the sample variance of past returns of the three assets during the most recent 24 months when the market is estimated to be in Regime 1, and finally, $\sigma^w(2)$ is the sample variance of past returns of the three assets during the most recent 24 months when the market is estimated to be in Regime 2.

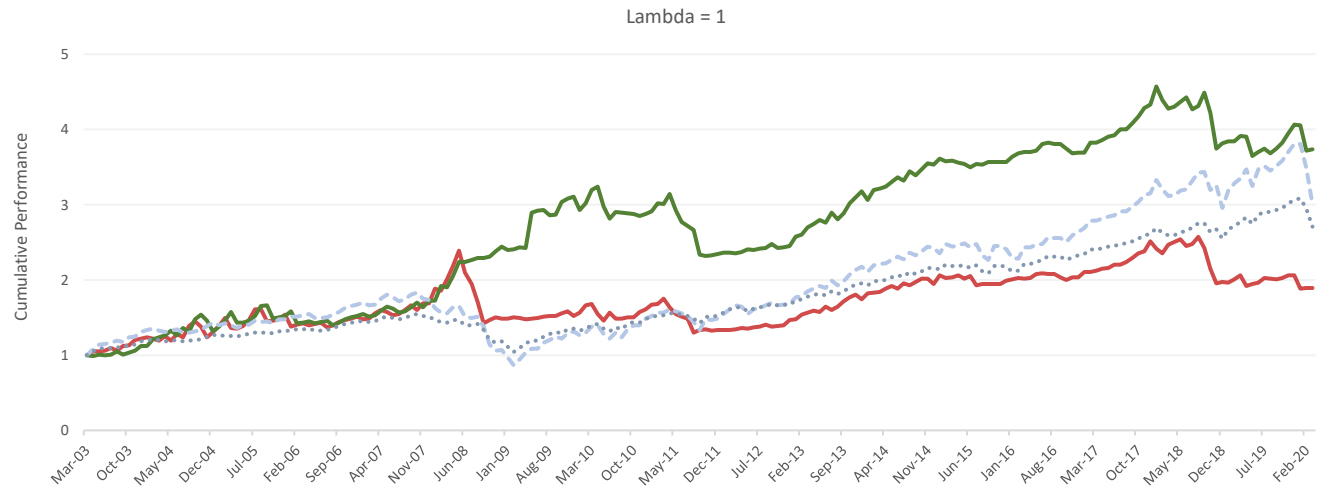
Since we are dealing with just three assets and have not included any risk-free asset in our portfolio, we will maintain restrictions on shorting. Thus, we will aim to build optimal long only portfolios with monthly rebalancing. We build portfolios based on the following objective function:

$$\max_x \mu^T x - \frac{1}{2}\lambda(x^T \Sigma x)$$

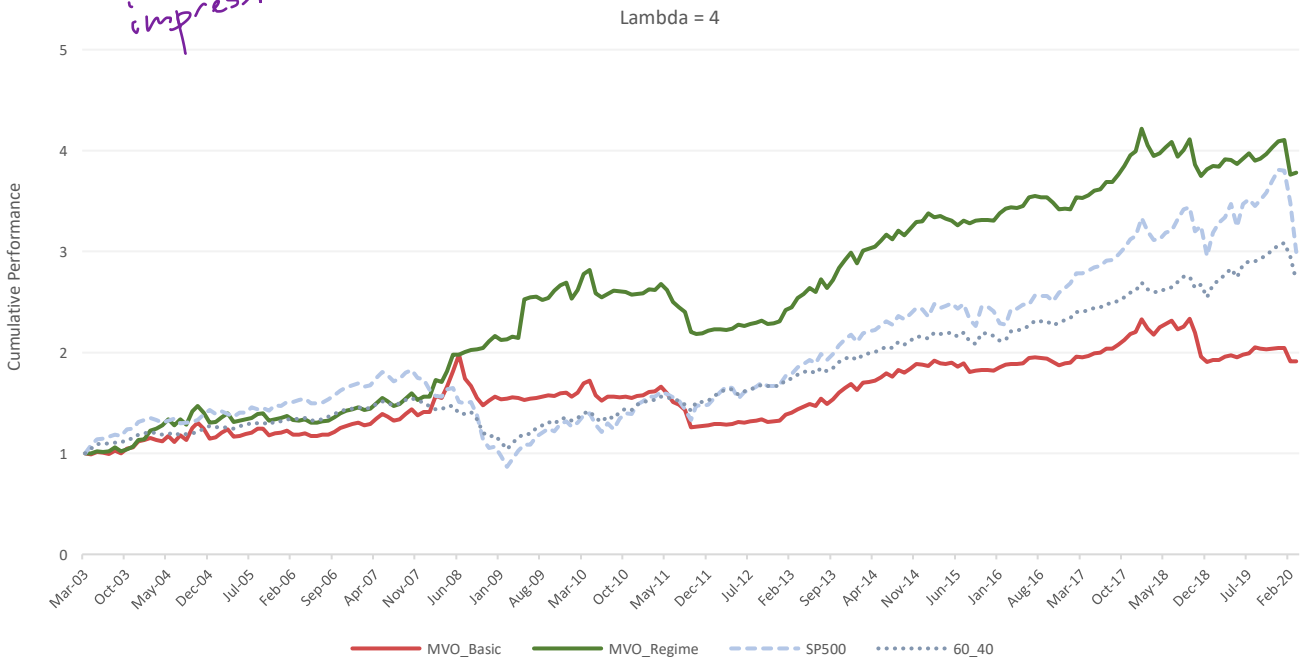
$$\text{s.t. } e^T x = 1$$

$$x_i \geq 0 \qquad \text{for } i = 1,2....N$$

Here, λ is the risk aversion parameter. The higher the λ, the more risk averse the investor is. And, e is a vector of 1s: the condition $e^T x = 1$ is just a mathematical notation for saying that the sum of weights allocated in the portfolio is 100%. We have tested the performance of the strategy with various values of λ and the results are quite consistent, in that the Regime based mean-variance optimized portfolios outperform the standard Markowtiz portfolio rebalanced monthly. The following charts show the performance of the strategy from 2003 to 2020 excluding transaction costs.

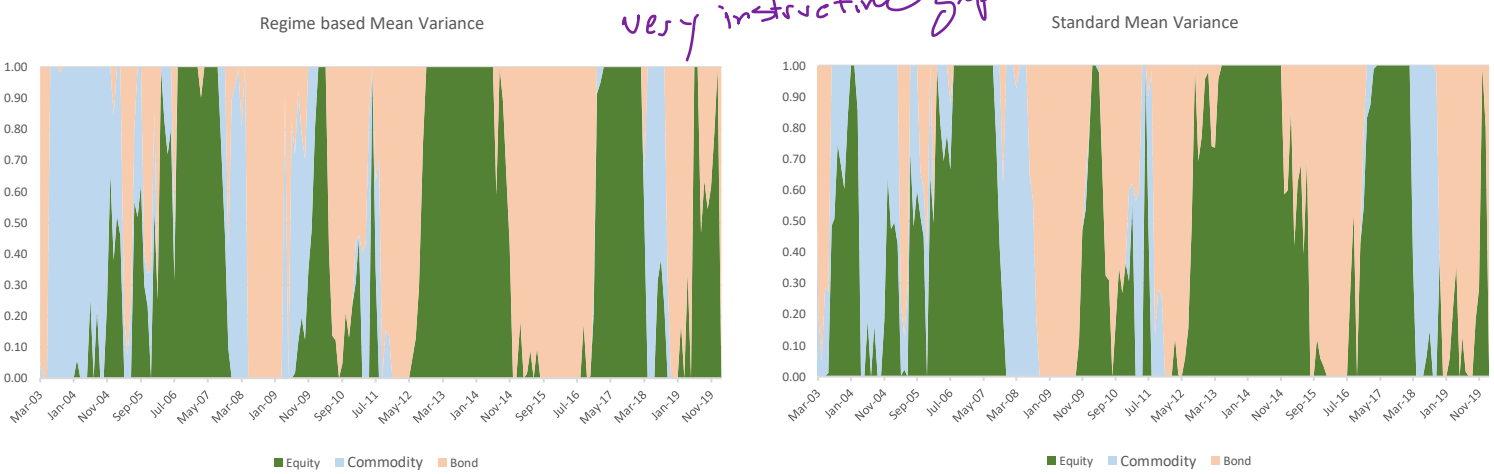


impressive!



The regime-based mean variance portfolio strategy has significantly outperformed the unconditional static mean-variance portfolio in the long run. Moreover, the strategy also outperforms the equity index and a comparable benchmark '60-40' which is 60% invested in equities and 40% invested in bonds. The outperformance can be alluded to the skill of the strategy in anticipating crises and market rally and re-allocating weights within the portfolio based on that. This can be verified by comparing the historical allocations in the three assets within both portfolios.

very instructive graphs



We can see that the regime-based strategy had rightly anticipated the market crisis of 2008 and reallocated its investments from commodities to bonds and thus the portfolio was shielded from large drawdowns, while the standard mean-variance portfolio suffered huge losses because of its allocations in commodities in the same time period. Likewise, in the year 2019, the regime-based portfolio anticipated a market rally and invested more than half of its wealth in equities that produced considerable gains. However, the standard mean variance portfolio was invested in bonds during the same period and lost out opportunity on those gains.

VII. Mean CVaR Portfolio Optimization

Conditional Value at Risk or CVaR is a prominent coherent risk measure that is commonly used in portfolio optimization techniques. It is also referred as the expected tail loss or expected shortfall. CVaR looks at the average loss that would be incurred if the VaR cutoff is exceeded.

Now, we denote a portfolio choice (weights given to the assets) by a vector x and the realization of random events, essentially a set of random returns of the assets, a vector y. Let $f(x, y)$ denote the loss function when we chose the portfolio x from a set X of feasible portfolios and y is the realization of a random event (a vector of asset returns of n assets). We denote $p(y)$ as the probability density function of y.

$$f(x, y) = -y^T x = -[y_1 x_1 + \ldots + y_n x_n]$$

The cumulative distribution of loss associated with the decision x can be represented as:

$$\psi(x,\gamma) = \int_{f(x,y)<\gamma} p(y)dy$$

From the definition of CVaR, substituting γ with $VaR_\alpha(x)$ CVaR with α% confidence level can be expressed as:

$$CVaR_\alpha(X) = \frac{1}{1-\alpha}\int_{f(x,y)\geq VaR_\alpha(X)} f(x,y)p(y)dy$$

Due to the presence of VaR in the equation it is going to be difficult to optimize this function. We instead look at the function below with certain desirable properties:

$$F_\alpha(x,\gamma) = \gamma + \frac{1}{1-\alpha}\int (f(x,y)-\gamma)^+ p(y)dy$$

Where $a^+ = \max\{a, 0\}$

1. $F_\alpha(x,\gamma)$ is a convex function of γ.
2. $VaR_\alpha(X)$ is the minimizer over γ of $F_\alpha(x,\gamma)$.
3. The minimum value over γ of the function $F_\alpha(x,\gamma)$ is $CVaR_\alpha(X)$.

As a consequence of these properties we can deduce that the set of weights x that minimize the $CVaR_\alpha$ of a portfolio can be obtained by minimizing $F_\alpha(x,\gamma)$ simultaneously over x and γ.

$$\min_x CVaR_\alpha(x) = \min_{x,\gamma} F_\alpha(x,\gamma)$$

We can discretize and do away with the probability distribution of scenarios of y, p(y) by choosing a set of scenarios y_i for $i = 1,\ldots,T$.

$$\bar{F_\alpha}(x,\gamma) = \gamma + \frac{1}{(1-\alpha)T}\sum_{i=1}^{T}(f(x,y_i)-\gamma)^+$$

Replacing Variance as a Risk measure in our definition of mean-variance optimization and including the equivalent risk measure of CVaR can be achieved by incorporating the above function as follows:

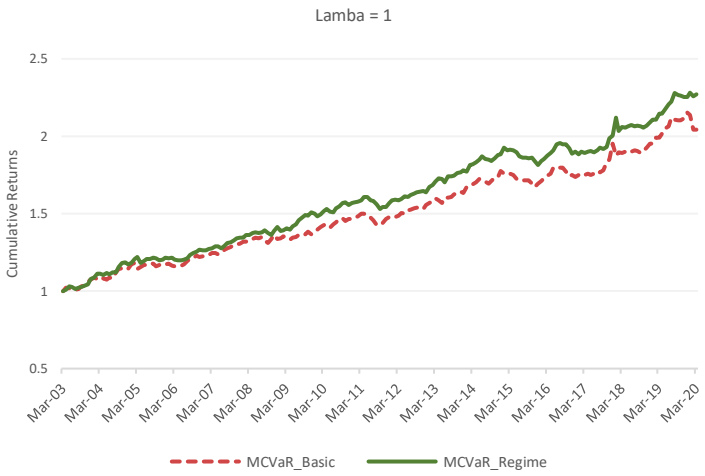
$$max_{z,\gamma,x}\ \mu^T x - \frac{1}{2}\lambda\left(\gamma + \frac{1}{(1-\alpha)T}\sum_{i=1}^{T} z_i\right)$$

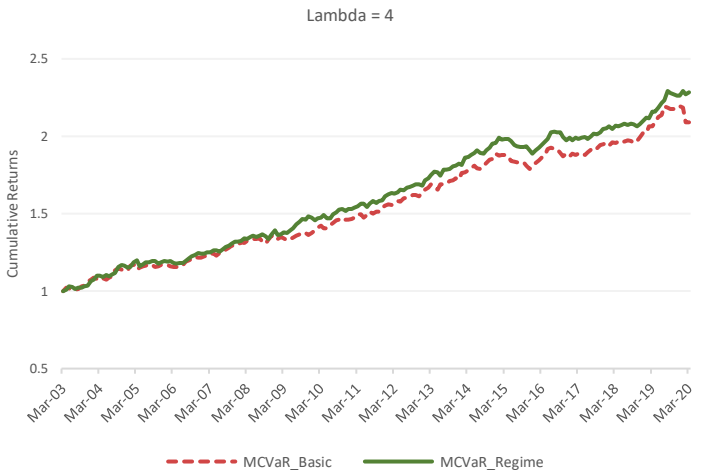
$$s.t.\ z_i \geq 0$$
$$z_i \geq f(x,y_i) - \gamma$$
$$e^T x = 1$$

Where $z_i = (f(x,y_i)-\gamma)^+$

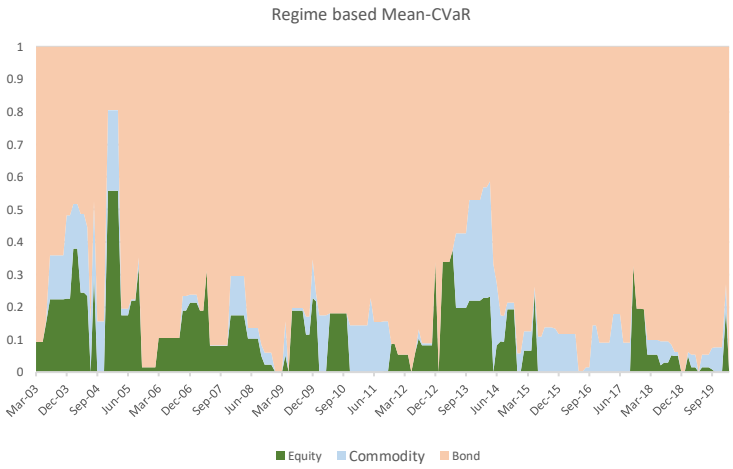
Again, λ can be thought of as a risk aversion parameter just like in the case of mean-variance optimization.

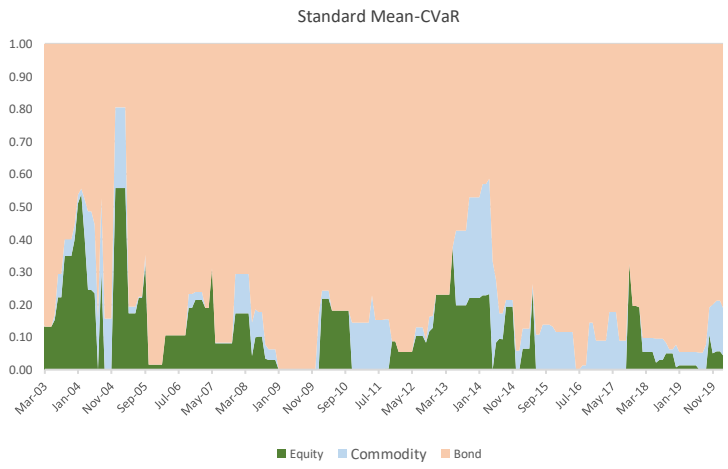
We implement the above optimization with various values of λ and an α value of 0.05 and compared the performance of regime-based Mean-CVaR portfolio optimization, taking into consideration scenarios of past 24 months, with a static unconditional standard Mean-CVaR portfolio optimization with similar assumptions. We see that regime based monthly rebalanced portfolio outperforms the static portfolio. This is consistent with our findings from Mean-Variance optimized portfolio.





To investigate where the outperformance comes from, we can look at the historical weights assigned to the three assets in the regime-based Mean-CVaR portfolio and compare it with the static unconditional portfolio.





An interesting finding is that the Mean-CVaR portfolio outperforms its static unconditional counterpart, however it underperforms the Mean-Variance portfolio and S&P500 index significantly. This is likely because of its large allocation to bonds most of the time which leads the portfolio to behave a lot like bonds in the long term. Since we have only 3 assets, we included a relatively small lookback period of 24 months, which may lead to representing only a small number of scenarios (with an α of 0.05) to compute the CVaR and thus the allocations are more conservative. However, the evidence that regime switching asset allocation significantly improves performance compared with an unconditional static alternative is interesting and a key finding of our research.

interesting!

Backtest Summary Statistics

	MVO_Regime	MVO_Standard	MCVaR_Regime	MCVaR_Standard
Start Date	April-03	April-03	April-03	April-03
Ann. Mean Returns	8.46%	4.47%	4.92%	4.43%
CAGR (since Inception)	8.14%	3.90%	4.96%	4.45%
CAGR (last 5 years)	2.14%	0.49%	2.62%	2.13%
Ann. Volatility	10.58%	11.34%	3.36%	3.52%
Ann. Sharpe Ratio	0.82	0.38	1.46	1.27
Information Ratio	0.75	0.39	0.64	0.48
Ann. Tracking Error	11.18%	11.36%	7.60%	9.05%
Maximum Drawdown	-22%	-36%	-4.90%	-5.20%

Note: Benchmark to calculate Tracking Error and Information Ratio is 60_40 portfolio (a portfolio invested in 60% Equities, 40% Bonds)

VIII. Conclusion

The project's point of focus was that different economic regimes require different asset allocations. In the presence of time varying investment opportunities, portfolio weights should be adjusted as new information arrives. Regime-switching models are well suited to capture the sudden changes of behavior and the phenomenon that the new dynamics of asset prices persist for several periods after a change. The asset classes considered were stocks, bonds, and commodities. The implementation of the Baum-Welch algorithm was an important contribution to the project that may be useful for future work on continuous time Hidden Markov Models. The estimation identified a call for an adaptive approach as the parameters of the estimated models were far from constant throughout the sample period. We finally constructed various efficient portfolios with different levels of risk and expected return based on Mean-Variance and Mean-CVaR portfolio optimization. Regime specific rebalance portfolios were shown to increase the return and at the same time reduce the risk – both the variance and maximum drawdown – compared to static unconditional portfolios optimized with similar objectives and constraints. The portfolios were evidence of achieving significantly higher Sharpe Ratios and CAGR out of sample. On closer analysis, we also found that the models accurately predicted the recessionary regime during current Covid-19 Financial Crisis and made reasonable adjustments to its allocations that protected the portfolio from suffering large drawdowns.

We believe this theme of research can be continued in future by including more number of assets in the portfolio and possibly evaluate the performance of these models on stock level data. It would also be relevant to do further research on continuous-time Hidden Markov Models and explore the behavior of more than two regimes of the economy. Our analysis was based on univariate gaussian models, so it makes sense to extend the complicity and employ multivariate gaussian mixture models to relate hidden states of the economy with micro and macro-economic indicators. A final suggestion for future work is to consider more smooth switching procedures for the portfolio adjustment for instance based on the confidence in the predicted state similar to the idea behind the Black-Litterman model.

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